

# Negative Exponent Law

---

**MA.8.NSO.1.3**

*I can rewrite a power with a negative exponent using a positive exponent, and evaluate it.*

*Yesterday the exponents stopped at 0.*

**Today they keep going — and  $5^{-2}$  is not a negative number.**

# Warm-Up

---

*On your own. Four minutes. No notes.*

1. Find the value:  $\frac{5^4 \cdot 5^0}{5^2}$
2. Which is greater,  $\sqrt{30}$  or 5.5?
3. Solve  $-3x = 21$
4. What is the reciprocal of  $\frac{3}{4}$  ? What is the opposite of  $\frac{3}{4}$  ?

# Warm-Up

---

*Answers.*

1. Find the value:  $\frac{5^4 \cdot 5^0}{5^2}$  **25**
2. Which is greater,  $\sqrt{30}$  or 5.5? **5.5**
3. Solve  $-3x = 21$   **$x = -7$**
4. What is the reciprocal of  $\frac{3}{4}$  ? What is the opposite of  $\frac{3}{4}$  ?  
 **$\frac{4}{3}$  and  $-\frac{3}{4}$**

# Notes

*Copy the table.*

## I. The pattern keeps going past zero

| $2^4$ | $2^3$ | $2^2$ | $2^1$ | $2^0$ | $2^{-1}$      | $2^{-2}$      | $2^{-3}$      |
|-------|-------|-------|-------|-------|---------------|---------------|---------------|
| 16    | 8     | 4     | 2     | 1     | $\frac{1}{2}$ | $\frac{1}{4}$ | $\frac{1}{8}$ |

$$2^{-1} = \frac{1}{2^1}$$

$$2^{-2} = \frac{1}{2^2} = \frac{1}{4}$$

$$2^{-3} = \frac{1}{2^3} = \frac{1}{8}$$

# Notes

---

## II. The negative exponent law

---

$$a^{-n} = \frac{1}{a^n} \qquad \frac{1}{a^{-n}} = a^n \qquad (a \neq 0)$$

$$5^{-2} = \frac{1}{5^2} = \frac{1}{25} \qquad \text{but} \qquad -5^2 = -25 \qquad \text{and} \qquad (-5)^2 = 25$$

**A negative exponent means reciprocal. It never means a negative value.**

The base keeps its own sign:  $(-2)^{-3} = \frac{1}{(-2)^3} = \frac{1}{-8} = -\frac{1}{8}$

# Notes

---

## III. The laws still work with negative exponents

---

$$6^{-2} \cdot 6^5 = \frac{1}{6^2} \cdot 6^5 = \frac{7776}{36} = 216$$

$$\text{or} \quad 6^{-2} \cdot 6^5 = 6^{-2+5} = 6^3 = 216$$

$$\frac{9^{-5}}{9^3} = 9^{-5-3} = 9^{-8} = \frac{1}{9^8}$$

**Add, subtract and multiply exponents exactly as before — the exponents are just integers now.**

# Example 1

---

*Negative exponents top and bottom.*

$$\frac{2^{-3}}{9^{-2}}$$

**Rewrite with positive exponents, then find the value.**

# Example 1

---

*Worked.*

$$\frac{2^{-3}}{9^{-2}} = \frac{9^2}{2^3} \quad \text{each power swaps sides and loses the minus}$$

$$\frac{9^2}{2^3} = \frac{81}{8} = 10.125 \quad \text{evaluate}$$

**Answer:  $81/8 = 10.125$**



# Your Turn

---

*Same steps, your numbers. Boards up when done.*

$$\frac{3^{-2}}{5^{-1}}$$

# Your Turn

---

*Answer.*

$$\frac{3^{-2}}{5^{-1}}$$

swap sides, positive exponents:  $5^1$  over  $3^2$

**Answer:**  $\frac{5}{9}$

# Example 2

---

*A quotient law crossing zero.*

$$\frac{7^0}{7^{-4}}$$

**Find the value.**

# Example 2

---

*Worked both ways.*

$$\frac{7^0}{7^{-4}} = 7^{0 - (-4)} = 7^4 \quad \text{quotient of powers: } 0 - (-4) = 4$$

$$\frac{7^0}{7^{-4}} = \frac{1}{7^{-4}} = 7^4 \quad \text{zero exponent, then move } 7^{-4} \text{ up}$$

$$7^4 = 2401 \quad \text{evaluate}$$

**Answer: 2,401**

# Your Turn

---

*Same steps, your numbers. Boards up when done.*

$$\frac{(-8)^{-4}}{(-8)^{-6}}$$

# Your Turn

---

*Answer.*

$$\frac{(-8)^{-4}}{(-8)^{-6}}$$

$$-4 - (-6) = 2, \text{ so } (-8)^2$$

**Answer: 64**

# Whiteboards • Question 1 of 9

---

*Boards up on three.*

$$8^{-3}$$

**Answer it.**

*Rewrite with a positive exponent.*

# Whiteboards • Question 1 of 9

---

*Answer.*

$$8^{-3}$$

reciprocal of  $8^3$

**Answer:**  $\frac{1}{8^3}$



# Whiteboards • Question 2 of 9

---

*Boards up on three.*

$$2^{-4}$$

**Answer it.**

*Find the value.*

# Whiteboards • Question 2 of 9

---

*Answer.*

$$2^{-4}$$

1 over  $2^4$

**Answer:**  $\frac{1}{16}$

# Whiteboards • Question 3 of 9

---

*Boards up on three.*

$$\frac{1}{10^{-2}}$$

**Answer it.**

*Rewrite with a positive exponent, then find the value.*

# Whiteboards • Question 3 of 9

---

*Answer.*

$$\frac{1}{10^{-2}}$$

the power moves up:  $10^2$

**Answer:  $10^2 = 100$**

# Whiteboards • Question 4 of 9

---

*Boards up on three.*

$$(-2)^{-3}$$

**Answer it.**

*Find the value.*

# Whiteboards • Question 4 of 9

---

*Answer.*

$$(-2)^{-3}$$

$$1 \text{ over } (-2)^3 = 1 \text{ over } -8$$

$$\textbf{Answer: } -\frac{1}{8}$$

# Whiteboards • Question 5 of 9

---

*Boards up on three.*

$$5^{-2}$$

Which is the value?

**A.**  $1/25$

**B.**  $-25$

**C.**  $-10$

**D.**  $1/10$

**Answer it.**

# Whiteboards • Question 5 of 9

---

*Answer.*

$$5^{-2}$$

Which is the value?

**A. 1/25**

**B. -25**

**C. -10**

**D. 1/10**

**Answer: A — 1/25**



# Whiteboards • Question 6 of 9

---

*Boards up on three.*

$$6^{-2} \cdot 6^5$$

Which expression is equivalent?

**A.**  $6^3$

**B.**  $6^{-7}$

**C.**  $6^{-10}$

**D.**  $36^3$

**Answer it.**

# Whiteboards · Question 6 of 9

---

*Answer.*

$$6^{-2} \cdot 6^5$$

Which expression is equivalent?

**A.  $6^3$**

**B.  $6^{-7}$**

**C.  $6^{-10}$**

**D.  $36^3$**

**Answer: A —  $6^3$**

# Whiteboards • Question 7 of 9

---

*Boards up on three.*

$$3^{-4} \cdot 3^4$$

**Answer it.**

*Find the value.*

# Whiteboards · Question 7 of 9

---

*Answer.*

$$3^{-4} \cdot 3^4$$
$$3^0$$

**Answer: 1**

# Whiteboards • Question 8 of 9

---

*Boards up on three.*

$$\frac{9^{-5}}{9^3}$$

**Answer it.**

*Write it as one power with a positive exponent.*

# Whiteboards • Question 8 of 9

---

*Answer.*

$$\frac{9^{-5}}{9^3}$$

$$9^{-8}$$

**Answer:**  $\frac{1}{9^8}$

# Whiteboards • Question 9 of 9

---

*Take your time. Boards up when you have written it.*

Ellen says the value of  $5^{-2}$  is  $-25$ .

Explain why she is wrong, and show the correct value.

**Write your answer in sentences.**

*Use the word reciprocal.*

# Whiteboards • Question 9 of 9

---

*Answer.*

Ellen says the value of  $5^{-2}$  is  $-25$ .

Explain why she is wrong, and show the correct value.

reciprocal, not opposite

**Answer:  $5^{-2} = 1/5^2 = 1/25$**



# IXL

---

*Last five minutes.*

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Stop at a SmartScore of 67 on each skill. Whatever you do not finish is tonight's practice.

## **Today's skills**

Understanding negative exponents — YBB

Evaluate powers with negative exponents — WGS